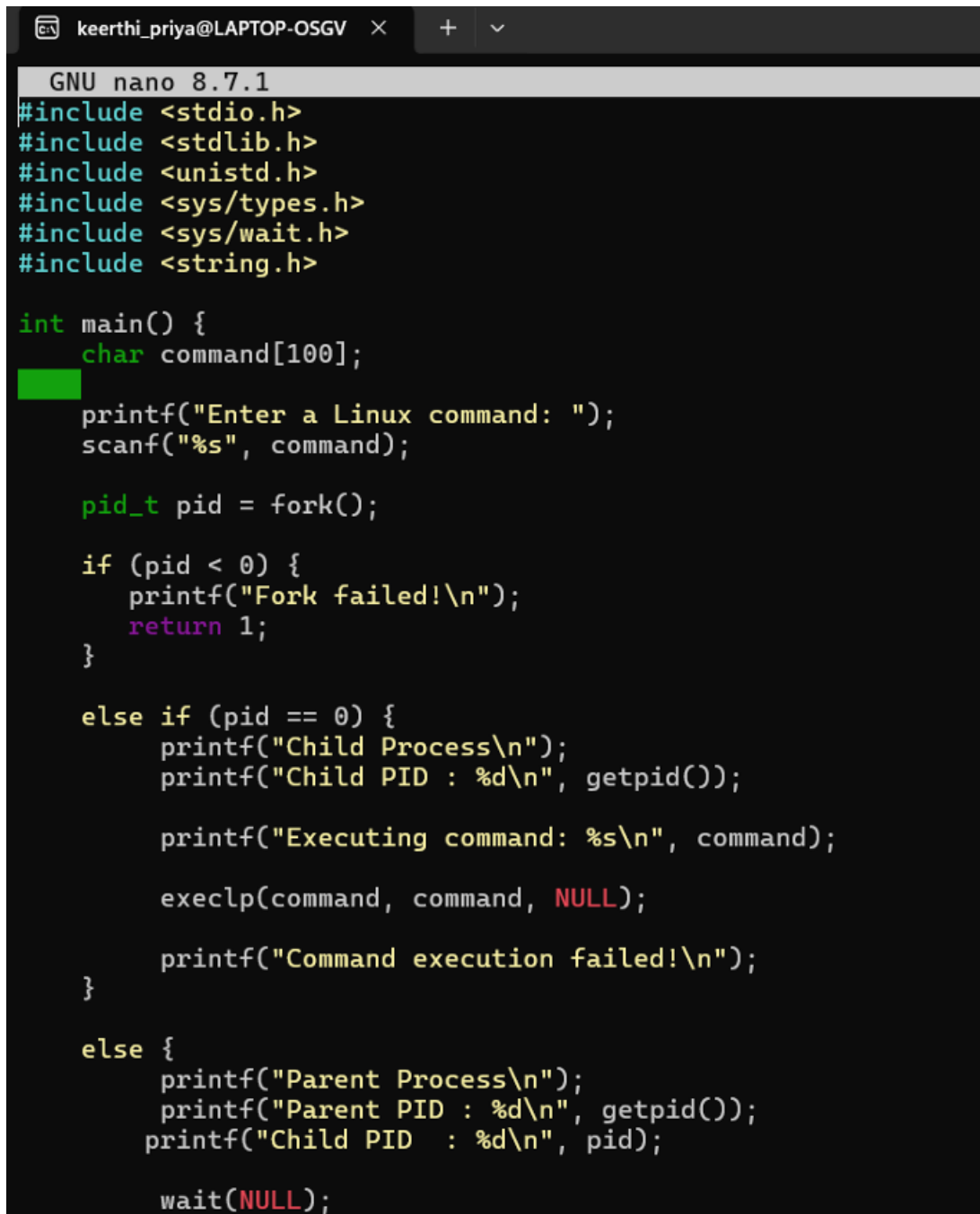


PRACTICAL WEEK 2

TASK 1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user Space and kernel space during execution.

A screenshot of a terminal window with a dark background. The title bar at the top shows a file icon, the username 'keerthi_priya@LAPTOP-OSGV', and window control buttons. The terminal content shows the GNU nano 8.7.1 editor with a C program. The program includes headers for stdio, stdlib, unistd, sys/types, sys/wait, and string. The main function prompts the user for a Linux command, forks a child process, and then uses execlp to execute the command in the child process. The parent process prints status information and waits for the child to complete.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);
    }
}
```

```
keerthi_priya@LAPTOP-OSGV × + v
GNU nano 8.7.1
int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
keerthi_priya@LAPTOP-OSGVCE7A:~$ mkdir OS_LAB
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd OS_LAB
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ nano task1.c
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ gcc task1.c -o task1
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 641
Child PID : 642
Child Process
Child PID : 642
Executing command: ls
task1 task1.c
Child process completed.
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ |
```

TASK 2: Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

```

Child process completed.
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ uname -a
Linux LAPTOP-OSGVCE7A 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               46 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      18
  On-line CPU(s) list:       0-17
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) Ultra 5 125H
CPU family:                  6
Model:                      170
Thread(s) per core:         2
Core(s) per socket:         9
Socket(s):                   1
Stepping:                    4
BogoMIPS:                    5990.40
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities
Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                      432 KiB (9 instances)
  L1i:                      576 KiB (9 instances)

```

```

  L1i:                      576 KiB (9 instances)
  L2:                      18 MiB (9 instances)
  L3:                      18 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-17
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Not affected
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                      Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected
  Old microcode:             Not affected
  Reg file data sampling:    Not affected
  Retbleed:                  Mitigation; Enhanced IBRS
  Spec rstack overflow:      Not affected
  Spec store bypass:         Mitigation; Speculative Store Bypass disabled v
                             ia prctl
  Spectre v1:                Mitigation; usercopy/swapgs barriers and __user
                             pointer sanitization
  Spectre v2:                Mitigation; Enhanced / Automatic IBRS; IBPB con
                             ditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_
                             S
  Srbds:                    Not affected
  Tsa:                      Not affected
  Tsx async abort:           Not affected
  Vmscape:                   Not affected
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ps
  PID TTY          TIME CMD
  346 pts/0    00:00:00 bash
  674 pts/0    00:00:00 ps
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1        0   0 16:08 ?        00:00:01 /sbin/init
root         2        1   0 16:08 hvc0     00:00:00 /init
root         6        2   0 16:08 hvc0     00:00:00 plan9 --control-socket 7
root        46        1   0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
systemd+    81        1   0 16:08 ?        00:00:00 /usr/lib/systemd/systemd

```

```
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1      0  0 16:08 ?        00:00:01 /sbin/init
root           2      1  0 16:08 hvc0     00:00:00 /init
root           6      2  0 16:08 hvc0     00:00:00 plan9 --control-socket 7
root          46      1  0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
systemd+      81      1  0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
root          92      1  0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
root         133      1  0 16:08 ?        00:00:00 /bin/sh /usr/lib/systemd
root         138      1  0 16:08 ?        00:00:00 /usr/sbin/cron -f -P
message+     143      1  0 16:08 ?        00:00:00 @dbus-daemon --system --
root         168      1  0 16:08 ?        00:00:00 /usr/bin/python3 /usr/bi
root         172      1  0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
syslog       192      1  0 16:08 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root         251      1  0 16:08 tty1     00:00:00 /usr/sbin/agetty --nores
_chrony      272     133  0 16:08 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony      283     272  0 16:08 ?        00:00:00 /usr/sbin/chronyd -n -F
root         293      1  0 16:08 ?        00:00:00 /usr/bin/python3 /usr/sh
root         344      2  0 16:08 ?        00:00:00 /init
root         345     344  0 16:08 ?        00:00:00 /init
keerthi+     346     345  0 16:08 pts/0    00:00:00 -bash
root         347      2  0 16:08 ?        00:00:00 login -- keerthi_priya
keerthi+     389      1  0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
keerthi+     393     389  0 16:08 ?        00:00:00 (sd-pam)
keerthi+     419     347  0 16:08 pts/1    00:00:00 -bash
keerthi+     684     346  0 16:26 pts/0    00:00:00 ps -ef
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ top
top - 16:27:57 up 19 min,  1 user,  load average: 0.02, 0.03, 0.00
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7651.1 total, 6970.3 free,  598.1 used,  234.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7053.1 avail Mem

   PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+    COMMAND
    1 root        20   0   24084   15360  11580 S   0.0   0.2   0:01.02  systemd
    2 root        20   0    3180    2208   2072 S   0.0   0.0   0:00.03  init-systemd(Ub
    6 root        20   0    3180    2048   1940 S   0.0   0.0   0:00.00  init
   46 root        19  -1   50424   16828  15556 S   0.0   0.2   0:00.23  systemd-journal
   81 systemd+    20   0   22416   14520  12028 S   0.0   0.2   0:00.06  systemd-resolve
   92 root        20   0   35440   12404   9236 S   0.0   0.2   0:00.45  systemd-udev
  133 root        20   0    2888    1948   1820 S   0.0   0.0   0:00.11  chronyd-starter
```

```
keerthi+     389      1  0 16:08 ?        00:00:00 /usr/lib/systemd/systemd
keerthi+     393     389  0 16:08 ?        00:00:00 (sd-pam)
keerthi+     419     347  0 16:08 pts/1    00:00:00 -bash
keerthi+     684     346  0 16:26 pts/0    00:00:00 ps -ef
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ top
top - 16:28:09 up 20 min,  1 user,  load average: 0.01, 0.03, 0.00
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7651.1 total, 6969.4 free,  598.9 used,  234.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7052.2 avail Mem

   PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+    COMMAND
    1 root        20   0   24084   15360  11580 S   0.0   0.2   0:01.02  systemd
    2 root        20   0    3180    2208   2072 S   0.0   0.0   0:00.03  init-systemd(Ub
    6 root        20   0    3180    2048   1940 S   0.0   0.0   0:00.00  init
   46 root        19  -1   50424   16828  15556 S   0.0   0.2   0:00.23  systemd-journal
   81 systemd+    20   0   22416   14520  12028 S   0.0   0.2   0:00.06  systemd-resolve
   92 root        20   0   35440   12404   9236 S   0.0   0.2   0:00.45  systemd-udev
  133 root        20   0    2888    1948   1820 S   0.0   0.0   0:00.11  chronyd-starter
  138 root        20   0    4472    3128   2880 S   0.0   0.0   0:00.01  cron
  143 message+    20   0    8788    5340   4664 S   0.0   0.1   0:00.06  dbus-daemon
  168 root        20   0   44096   29780  14636 S   0.0   0.4   0:00.31  networkd-dispat
  172 root        20   0   18432    9384   8148 S   0.0   0.1   0:00.17  systemd-logind
  192 syslog      20   0   220548   5864   4624 S   0.0   0.1   0:00.16  rsyslogd
  251 root        20   0    5492    2712   2524 S   0.0   0.0   0:00.02  atetty
  272 _chrony      20   0   23564   11196   7220 S   0.0   0.1   0:00.14  chronyd
  283 _chrony      20   0   12096    2340   1696 S   0.0   0.0   0:00.01  chronyd
  293 root        20   0   123460   32708  18100 S   0.0   0.4   0:00.25  unattended-upgr
  344 root        20   0    3184    1112    980 S   0.0   0.0   0:00.00  SessionLeader
  345 root        20   0    3200    1252   1108 S   0.0   0.0   0:00.34  Relay(346)
  346 keerthi+    20   0    6268    5600   3928 S   0.0   0.1   0:00.09  bash
  347 root        20   0    8648    5536   4692 S   0.0   0.1   0:00.01  login
  389 keerthi+    20   0   22340   13624  11000 S   0.0   0.2   0:00.16  systemd
  393 keerthi+    20   0   22916    4088   2088 S   0.0   0.1   0:00.00  (sd-pam)
  419 keerthi+    20   0    6136    5416   3864 S   0.0   0.1   0:00.02  bash
  694 keerthi+    20   0    8172    6008   3900 R   0.0   0.1   0:00.05  top
```

